ASRC Fuel Tank Robot Simulation Review

M module backbone studies, 8 November 2026

The 6 November review identified two blocking problems for the 63 mm PA12 and NiTi front: active straightening needed far more than the 5% axial strain diagnostic, and no low-strain configuration could hold the 500 g head near the 20 degree working elevation. Today's work traced both to the M module and tested a specific fix in simulation.

Summary of today's work

- **M stiffness requirement sweep.** Scaling only the M module EA and EI for U4 plus M2 shows that axial stiffness alone leaves a 7.1 degree tangent error floor, set by M bending stiffness between anchor disks. The first passing combination is EA x5 (about 830 N) with EI x1.25 (about 0.055 N m^2), anchors [4,5,6], confirmed at 16 cells per module.
- **NiTi backbone concept.** A central 1.8 mm NiTi rod in each M module meets both requirements: EA rises to about 102 to 191 kN and EI to 0.065 to 0.083 N m^2 (E 40 to 75 GPa). Rod mass, 1.58 g per module, is included.
- **Straightening with the backbone.** U4 plus M2 passes every straightness check at 15, 20 and 25 degrees, with about 1 to 2 mm offset and about 0.1% strain. U6 plus M3 and U8 plus M4 still fail with three or fewer anchor sets, including an exhaustive layout search.
- **Steering with the backbone.** At 20 degrees, U4 plus M2 reaches the 180 degree yaw search cap with world head elevation held within 20 plus or minus 5 degrees, confirmed at 8 cells per module. At 15 degrees, 165 degrees is confirmed. The 6 November design found no such state.
- **Test plan.** A build and test plan for the backbone M module defines specimens, guide spacing, tests and pass criteria.

Item	6 November baseline	With 1.8 mm NiTi backbone (simulation)
U4 plus M2 straightening at 20 deg, at or below 5% strain	94.7 mm offset, 61.3 deg, fails	1.7 mm offset, 4.0 deg, 0.10% strain, passes
Same, 15 and 25 deg	Fails	Passes at both
Yaw with head held at 20 plus or minus 5 deg	No solution found	180 deg (search cap), 46 to 98 N peak tendon
U6 plus M3, U8 plus M4 straightening	Fail	Still fail with up to 3 anchor sets
Physical evidence	Database properties	Rod properties calculated only; tests not yet run

Main engineering decision
Adopt a central NiTi backbone in the M modules as the lead design for U4 plus M2, and build and test the backbone M module before further simulation refinement. Keep U4 plus M2 as the only development length; longer fronts need additional anchor sets inside M or a different M design.

Why the current M module fails

The supplied M module has EA 166.7 N and EI 0.0441 N m². The U module EA is about 770 times higher. Tendons that lift the head compress the M modules: at 5% strain M carries only about 8.3 N, while the head and module gravity moment needs roughly 39 N of tendon tension for U4 plus M2. This was the priority 1 problem in the 6 November review.

M stiffness requirement sweep

The 6 November planar kernel was reused unchanged. Only M EA (x1 to x1000) and M EI (x1 to x5) were scaled, for U4 plus M2 at 20 degrees with the 500 g head and the same four anchor layouts, optimizer, seeds and pass criteria. The unscaled case reproduces the reported 6 November result (94.68 mm at 8 cells against 94.77 mm at 16 cells). The full grid of 78 combinations runs in about 2 minutes.

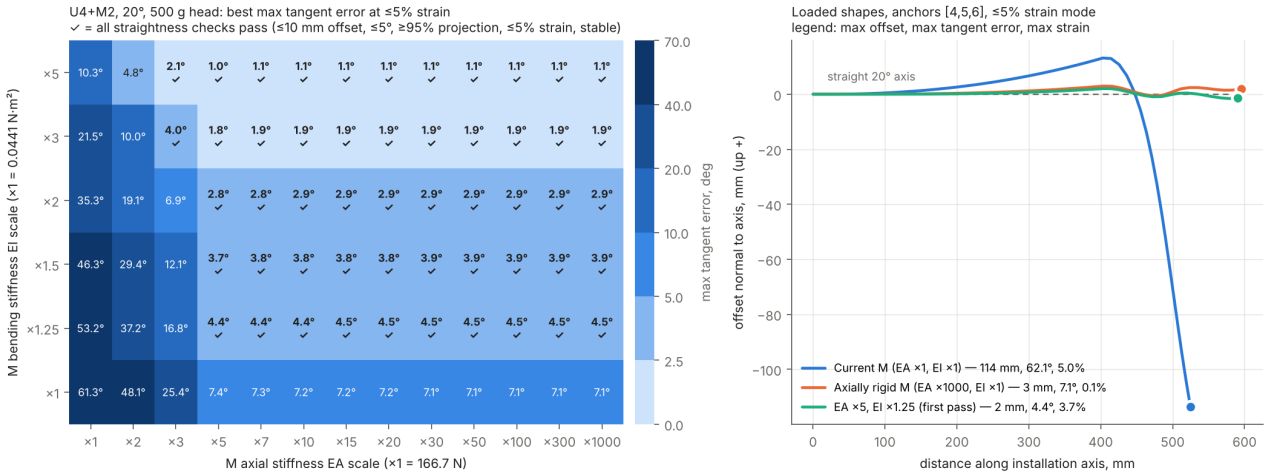


Figure 1 Left: best maximum tangent error at or below 5% strain over the EA and EI grid; ticks mark combinations passing all straightness checks. Right: loaded shapes with anchors [4,5,6].

Raising EA alone straightens the front to about 3 mm offset, but the maximum tangent error stops at about 7.1 degrees even at EA x1000. Between anchor disks the head weight produces a bending moment that varies linearly along each 96 mm M module, while the tendons apply a constant moment over that span. The residual rotation, about ΔM L / (8 EI) = 0.44 x 0.096 / (8 x 0.044) ≈ 0.12 rad (6.9 degrees), is set by M bending stiffness and matches the solver.

Threshold cases re-solved at 16 cells per module (anchors [4,5,6])

EI scale	EA scale	Offset mm (16 cell)	Tangent deg (16 cell)	Strain (16 cell)	All checks
x1	x1000	3.01	7.28	0.11%	Fail
x1.25	x3	14.51	16.85	4.86%	Fail
x1.25	x5	2.07	4.29	3.73%	Pass
x1.5	x5	1.61	3.71	3.74%	Pass
x2	x5	1.04	2.79	3.74%	Pass
x3	x3	8.66	4.01	4.87%	Pass
x5	x3	4.62	2.13	4.86%	Pass

Every 8-cell pass or fail classification was unchanged at 16 cells; the largest tangent change was 0.09 degrees.

Backbone concept and sizing

A central superelastic NiTi rod, bonded to the M module end disks and guided laterally inside the module, adds axial and bending stiffness in parallel with the PA12 structure. It does not change the 24 mm tendon moment arm. Two moduli bracket the uncertain NiTi phase state: 40 GPa (martensite or low-modulus) and 75 GPa (austenite).

Property	M (database)	M + rod, E 40 GPa	M + rod, E 75 GPa	Required (sweep)
EA, N	166.7	101,954	191,018	833 or more
EI, N m² (both planes)	0.0441	0.0647	0.0828	0.055 or more
Module torsion, N m/rad	0.437	0.599	0.740	Not limiting
Mass per M module, g	23.948	25.524	25.524	—

Rod diameter options

Rod diameter	Added EA, N	Added EI, N m²	M EI scale	Max guide spacing for 248 N	Surface strain, 180 deg over 192 mm
1.5 mm	71k to 133k	0.010 to 0.019	x1.23 to x1.42	20 to 27 mm	1.2%
1.8 mm (selected)	102k to 191k	0.021 to 0.039	x1.47 to x1.88	29 to 39 mm	1.5%
2.0 mm	126k to 236k	0.031 to 0.059	x1.71 to x2.34	35 to 48 mm	1.6%

Ranges are for E 40 to 75 GPa. Guide spacing uses pinned Euler buckling of the rod carrying twice the 124 N sum of the passing straightening tensions.

Engineering consequences

- 1.5 mm is marginal at the lower modulus (EI x1.23 against a x1.25 requirement). 1.8 mm passes with margin at both moduli.
- Intermediate rod guides are needed. A 24 mm spacing gives 353 N (E 40 GPa) and 662 N (E 75 GPa) Euler loads against a 248 N design load.
- Both rod ends must carry axial load into the anchor disks without slip; this joint is a new critical component.
- A stiffer M needs more tendon force to bend. The steering study below includes this effect.
- Rod end fittings and guides are not yet included in the mass. Friction, NiTi transformation hysteresis and fatigue are not modelled.

Active straightening with the backbone

The 6 November planar study was repeated for all three configurations at 15, 20 and 25 degrees, with the rod EA, EI and mass added to every M module. Values are the best stable result at or below 5% strain at 8 cells per module. Pass means offset at most 10 mm, tangent error at most 5 degrees, axial projection at least 95%, strain at most 5% and local stability with 1 m of extra rope.

Configuration	Pitch	Current M	Rod, E 40 GPa	Rod, E 75 GPa
U4 plus M2	15 deg	Fail: 89.7 mm, 57.5 deg, 4.96%	**Pass**: 1.8 mm, 4.1 deg, 0.10%	**Pass**: 1.2 mm, 3.2 deg, 0.09%
U4 plus M2	20 deg	Fail: 94.7 mm, 61.3 deg, 4.96%	**Pass**: 1.7 mm, 4.0 deg, 0.10%	**Pass**: 1.2 mm, 3.1 deg, 0.09%
U4 plus M2	25 deg	Fail: 99.1 mm, 64.9 deg, 4.96%	**Pass**: 1.7 mm, 3.7 deg, 0.09%	**Pass**: 1.2 mm, 3.0 deg, 0.09%
U6 plus M3	15 deg	Fail: 174.5 mm, 82.5 deg, 4.95%	No stable result	Fail: 6.6 mm, 12.9 deg, 0.12%
U6 plus M3	20 deg	Fail: 176.8 mm, 87.2 deg, 4.95%	Fail: 10.5 mm, 15.1 deg, 0.12%	Fail: 9.6 mm, 10.4 deg, 0.12%
U6 plus M3	25 deg	Fail: 179.4 mm, 91.9 deg, 4.95%	Fail: 8.4 mm, 15.2 deg, 0.11%	Fail: 6.6 mm, 11.0 deg, 0.11%
U8 plus M4	15 deg	Fail: 281.4 mm, 93.2 deg, 4.95%	Fail: 162.3 mm, 89.2 deg, 0.25%	Fail: 23.6 mm, 30.8 deg, 0.21%
U8 plus M4	20 deg	Fail: 278.0 mm, 98.1 deg, 4.95%	Fail: 135.2 mm, 102.5 deg, 0.28%	Fail: 19.2 mm, 29.8 deg, 0.21%
U8 plus M4	25 deg	Fail: 308.5 mm, 103.1 deg, 4.96%	Fail: 101.6 mm, 111.1 deg, 0.32%	Fail: 18.0 mm, 28.3 deg, 0.21%

- **U4 plus M2 passes at every pitch and both moduli**, always with anchors [4,5,6]. At 20 degrees and E 40 GPa the top-rope tensions are 91.5, 21.4, 8.5 N.
- **U6 plus M3 and U8 plus M4 do not pass.** Every three-set layout with one set at the head was searched at 20 degrees for both moduli (28 layouts for U6 plus M3 and 55 for U8 plus M4). The closest case, U6 plus M3 with anchors [7,8,9], reached 5.04 degrees tangent error but 24.5 mm offset. The longer M sections leave longer spans between anchor disks, which the 1.8 mm rod cannot hold straight.
- The no-rod column reproduces the 6 November values, for example 94.68 mm at 20 degrees for U4 plus M2.

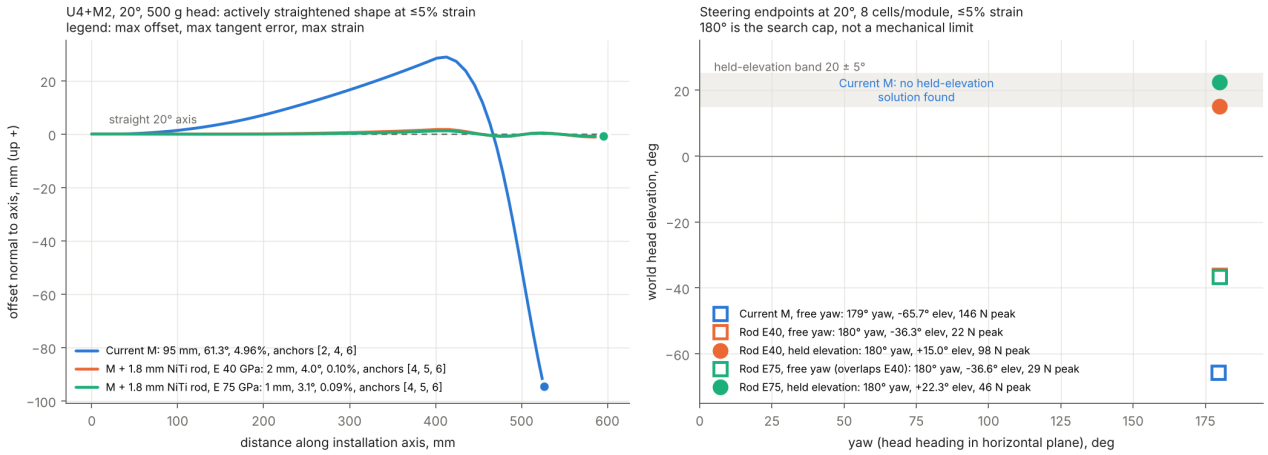


Figure 2 Left: U4 plus M2 actively straightened at 20 degrees, without and with the rod. Right: steering endpoints at 20 degrees, 8 cells per module; filled markers hold world head elevation within 20 plus or minus 5 degrees.

Spatial steering and loaded head recovery

The 6 November spatial steering study was repeated for U4 plus M2 with the backbone, using the same rod kernel, bounds and acceptance rules: 0 to 250 N per rope, coarse strain bound 4.75%, 180 degree search cap, geometry, capsule and fixed-winch stability checks. Candidates are refined from 2 to 4 to 8 cells per module, with re-optimization on the finer mesh when needed. Only states passing at 8 cells are reported. Held elevation means yaw with world head elevation within base pitch plus or minus 5 degrees.

Variant	Pitch	Free yaw (elevation, peak N)	Yaw with held elevation (elevation, peak N)	Up / down
Current M	20 deg	179.4 deg (-65.7, 146.5 N)	None found	-56.6 / 113.8
Rod, E 40 GPa	15 deg	180.0* deg (-33.3, 24.4 N)	**165† deg** (+10.0, 56.5 N)	180.0* / 180.0*
Rod, E 40 GPa	20 deg	180.0* deg (-36.3, 22.2 N)	**180* deg** (+15.0, 98.3 N)	180.0* / 180.0*
Rod, E 40 GPa	25 deg	180.0* deg (-39.5, 21.1 N)	**180* deg** (+30.0, 246.9 N)	180.0* / 180.0*
Rod, E 75 GPa	20 deg	180.0* deg (-36.6, 29.3 N)	**180* deg** (+22.3, 46.2 N)	180.0* / 180.0*

** Reaches the 180 degree search cap; not a demonstrated mechanical limit. † At 15 degrees no 180 degree held state was stable at 8 cells; 165 degrees is the largest lower cap that passed. Up and down are projected turning angles relative to the base.*

- **The rod resolves the loaded head recovery problem in simulation.** The current M finds no held-elevation state; with the rod, the head is swept through the full 180 degree search range while held near the working elevation, at well under 1% strain.
- Strain no longer limits the result (at most 0.62%). The active limits are now tendon force, stability and the elevation band.
- The no-rod control reproduces the 6 November values exactly: 179.42 degrees, minus 65.67 degrees elevation, 146.48 N.

Tendon forces for the held-elevation endpoints

Case	Anchors	Tensions per set [up, right, down, left], N
E 40 GPa, 15 deg, 165 deg	[4, 5, 6]	[1.3, 56.5, 9.7, 0.0] ; [1.3, 0.0, 0.5, 56.0] ; [26.4, 30.0, 4.8, 5.2]
E 40 GPa, 20 deg, 180 deg	[4, 6]	[0.0, 80.3, 98.3, 76.6] ; [38.6, 6.5, 31.6, 4.2]
E 40 GPa, 25 deg, 180 deg	[4, 6]	[0.5, 238.1, 246.9, 231.7] ; [8.3, 27.6, 1.8, 35.3]
E 75 GPa, 20 deg, 180 deg	[4, 6]	[0.0, 1.1, 11.0, 0.3] ; [0.4, 37.8, 1.0, 46.2]

These are one allocation found by the optimizer, not minimum-force solutions. The 8-cell re-optimization maximizes angle only, so tensions can rise during refinement: at 20 degrees and E 40 GPa the coarse candidate needed about 33 N peak, the 8-cell state 98 N. The 25 degree case sits near the 250 N bound and at the edge of its elevation band. Whether intermediate postures form a continuous controllable path has not been established.

Numerical verification

- All kernels are the 6 November planar and spatial rod models, reused without modification; only module properties change.
- Controls without the rod reproduce the 6 November planar (94.68 mm) and spatial (179.42 degrees, minus 65.67 degrees, 146.48 N) results.
- Spatial model with the rod: energy gradient against finite differences within 1.4e-9, and planar-to-spatial energy agreement within 5e-15.
- EA and EI sweep: all threshold classifications unchanged from 8 to 16 cells; largest tangent change 0.09 degrees.
- Steering: every reported state passes all acceptance rules at 8 cells. The 15 degree held case required a lower angle cap (165 degrees).
- Not modelled: friction, contact, NiTi transformation and hysteresis, printed part buckling, fatigue, rod end fittings, tank geometry.

Build and test plan for the backbone M module

Full details are in M_BACKBONE_TEST_PLAN.md in the backbone study folder. Specimens are the current M module as a control, the M module with a 1.8 mm superelastic NiTi rod and three intermediate guides at 24 mm spacing, and a two-module M2 chain.

Test	Method	Pass criterion
Rod material	Tension to 6% and unload; Af by DSC or bend and free recovery	Moduli and Af recorded; choose 40 or 75 GPa case
Axial stiffness	Plus and minus 150 N, 5 cycles, LVDT or DIC across end disks (5 µm resolution)	EA at least 833 N; no joint slip
Bending stiffness	Cantilever, 0.1 to 0.6 N tip load, three orientations	EI at least 0.055 N m² in every plane
Rod buckling	Compress via rod end; guide spacings 20, 24, 32 mm	No buckling at or below 248 N
Torsion (optional)	Plus and minus 10 deg twist	Recorded, expected 0.60 to 0.74 N m/rad
Tendon-driven bending	M2 chain, one rope 0 to 60 N, camera	Curvature within 15% of model with measured EI
Bending fatigue	180 deg bend over M2, 10,000 cycles	EI change under 10%; no slip or cracking
Integrated U4 plus M2	20 deg, 500 g dummy head, anchors [4,5,6], 91.5 / 21.4 / 8.5 N	Offset at most 10 mm; tangent at most 5 deg

Measured values should be added to a copy of module_stiffness.sqlite3 as a new module entry and the backbone studies re-run with it.

Problems and next decisions

Priority	Problem	Required work
1	Backbone M module validation	Build and run the test plan. The bending stiffness, rod end joint and buckling at the guide spacing decide whether the simulated results hold.
2	Tendon force allocation	Search for minimum-force tension sets at held elevation, and check continuous paths between postures. Reduce the 98 N and 247 N peak cases.
3	Entry and storage	Re-run the front-only tank entry screen with the new low-strain straight shape; add sensor CAD, cable envelopes, sleeve contact and telescopic storage for the 63 mm front.
4	Rear mechanism	Select stage sizes, overlap, antirotation guides, drive and locks, using the new front reactions.
5	Integrated validation	Full three dimensional tank sweep with real head dimensions, then unloaded and incremental payload prototypes.
6	Longer fronts (optional)	U6 plus M3 and U8 plus M4 need more anchor sets inside M or a stiffer M; revisit only if U4 plus M2 reach is insufficient.

Recommendation

Proceed with U4 plus M2, three or fewer anchor sets, and a central 1.8 mm NiTi backbone in each M module. The simulations now show straightening within the strain diagnostic at 15 to 25 degrees and full yaw at the working elevation, but every result rests on calculated rod properties. Physical module tests are the next gate. If measured bending stiffness falls below about 0.055 N m², increase the rod diameter to 2.0 mm or shorten the unsupported M spans before changing the architecture.

The 5% strain value, 250 N tendon bound and 180 degree angle cap remain numerical diagnostics, not hardware ratings.

Sources and reproduction

Study folders: ASRC_kevin/telescopic_planar_study/m_axial_sweep_20261108 and backbone_20261108, each with a README, scripts and saved results. This report is generated by ASRC_kevin/telescopic_planar_study/reports/build_report_2026_11_08.py.